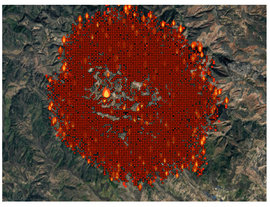
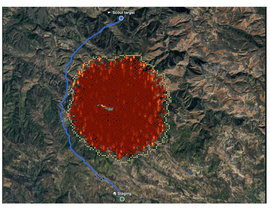
# How the dog decides



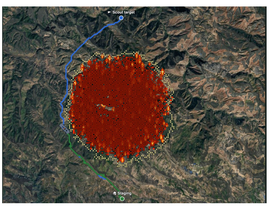
## 1. Predict

Forecast when each 50 m cell will burn, in 15-minute steps.



## 2. Plan a safe route

Pass every spot 30+ min before the fire. Brush (0.5×) and forest (0.7×) are slower; uphill is slowest. No water or slopes over 35°.



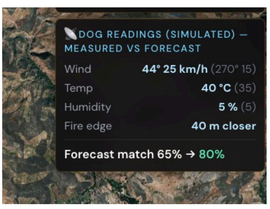
## 3. Keep an escape path

Aim for a way home that stays clear for 60 min, re-planned as the dog moves.



## 4. Scout

Walk to the fire edge and measure wind, temperature, humidity and where the edge really is.




## 5. Feed back

The readings correct the forecast, and the loop starts again.

Simulator views on real terrain. Robot image: design concept; sensors are simulated.

# How we tested it

Two fires run at once on the same terrain:

**Forecast fire**
What the dog plans on.

**True fire (hidden)**
Hotter, drier, with a wind shift. The dog only learns about it through its readings.

Mock simulation; no physical robot yet.

# Status and next steps

- ✓ ML fire forecast trained and tested
- ✓ 3D browser simulator on real terrain
- ✓ Robot-dog loop in mock simulation
- ○ Forecast + planner on the dog's controller
- ○ Real sensors feeding the forecast
- ○ Tests on real robot dogs

# References

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections); Open-Meteo weather archive; ESA WorldCover 10 m land cover; Cesium World Terrain; Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.

Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*